# Beware of the 'moving target' – uninvolved limb strength increases to exceed pre-operative values during rehabilitation after ACL reconstruction in male professional and recreational athletes

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**ABSTRACT**

**Objective**: To describe the changes in quadriceps and hamstrings muscle strength in the uninvolved limb of male professional and recreational athletes during rehabilitation after ACLR and compare to preoperative strength values.

**Design**: Prospective longitudinal study.

**Methods**: During rehabilitation, 665 participants who underwent unilateral ACLR performed a strength test preoperatively and every 6 weeks after surgery for up to 9 months. Isokinetic quadriceps and hamstrings strength of the uninvolved limb were measured at an angular velocity of 60°/s and normalized to body weight (Nm/kg).

**Results**: Quadriceps and hamstrings strength of the uninvolved limb gradually increased during rehabilitation until 6 months post-ACLR, and plateaued thereafter. Postoperative quadriceps' strength significantly exceeded preoperative values by 3 months for professional (2.99 Nm/kg, CI 95% 2.93-3.04, p=0.007) and by 6 months for recreational athletes (2.77Nm/kg, CI 95% 2.71-2.83, p<0.001), and hamstrings exceeded pre-surgery levels by 4.5 months (1.71Nm/kg, CI 95% 1.67-1.74, p<0.001 and 1.43 Nm/kg, CI 95% 1.40-1.46, p=0.002, professional and recreational athletes respectively).

**Conclusion**: In male professional and recreational athletes, uninvolved limb quadriceps and hamstrings strength gradually improved after ACLR, until 6 months after surgery. The uninvolved limb's strength may present as a 'moving target' that requires consistent monitoring during rehabilitation.

***Key words****: ACL reconstruction, uninvolved limb, isokinetic muscle strength, rehabilitation*

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## INTRODUCTION

After anterior cruciate ligament reconstruction (ACLR), residual impairments in knee muscle strength are pronounced in the first 6 months after surgery. The impairments may persist for over 2 years which may contribute to decreased performance at return to sport (RTS).[29,33] Reduced knee strength is suggested as one of the outcomes related to reinjury rates,[7,19] which are documented to be 5-9% in the ipsilateral leg, and 8-26% in the contralateral leg,[24,27,28,44,46] with higher rates in younger athletes.[42] Recovery of quadriceps and hamstrings strength is consequently one of the most used criteria to determine readiness to RTS after ACLR.[3]

The Limb Symmetry Index (LSI) is commonly used as a metric to evaluate muscle strength recovery, with the uninvolved leg as reference. In the final stage of rehabilitation a LSI ≥90% for quadriceps and hamstrings strength has been recommended.[40] However, the use of LSI to track strength progression after ACLR has been questioned. Some authors suggest that LSI using postoperative values of the uninvolved limb strength is insufficient to ensure adequate strength in the operated limb and may overestimate its recovery. Therefore, it has been recommended to compare the involved limb's strength to preoperative levels of the uninvolved limb, as the estimated preinjury knee extensor strength and preoperative LSI seem more sensitive metrics to identify those at higher risk of a secondary injury post-ACLR.[43,47]

The evidence regarding changes in the uninvolved limb muscle strength during rehabilitation is inconsistent. Some studies report no changes in strength through rehabilitation when measured at multiple time points[4,10,30,34] and other studies report lower postoperative strength in the uninvolved limb compared to matched healthy

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controls, 1 to 4 years after surgery.[13,14,26,38] Currently, there are no studies reporting consistent monitoring of uninvolved limb strength from preoperative to postoperative that can provide a reference for rehabilitation progression.

The purpose of this study was to describe the progression of the uninvolved limb's quadriceps and hamstrings strength throughout rehabilitation after surgery, in professional and recreational male athletes. Additionally, it was sought to determine the uninvolved limb's strength at different time points of rehabilitation compared to preoperative levels. It was hypothesized that with sufficient training stimulus the strength of the uninvolved limb will not decline during the rehabilitation period compared to preoperative values.

## METHODS

### Study design and participants

This prospective longitudinal study consecutively included patients, from Aspetar Hospital's ACL registry, after the first sports physician or surgeon consultation. Participants were male, aged 15 to 55 years, who scored ≥5 on the Tegner Activity Scale, and without a current or previous contralateral knee injury, ACL injury or ACLR. Female athletes with ACL injury were not included in this study as they represented approximately 4% of the clinical cohort and thus, the analyses adjusting for sex was underpowered and unable to be adequately performed. Patients sustained an ACL injury and subsequently underwent reconstruction, with either patellar, hamstrings or quadriceps tendon, or allograft, and evidence-based rehabilitation[20,40] between January 2017 and October 2022. Participants were further excluded: if a graft from the

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contralateral knee was used for the reconstruction, if their rehabilitation program or follow-up assessments were compromised by COVID-19 restrictions, or if they showed compliance issues such as patients who stopped rehabilitation for more than six weeks or stopped rehabilitation altogether, with fewer than two assessment follow-ups (**FIGURE 1**).

The study protocol was approved by the local institutional review board (Anti-Doping Lab Qatar – F2017000227) and all participants provided written informed consent.

**Rehabilitation Program**

Participants undertook a supervised, standardized, evidence-based rehabilitation program that was divided into criteria-based progression phases: early, intermediate, and advanced.[20,40] An additional sports specific training phase was included for professional athletes or recreational athletes involved in pivoting sports. The early phase of rehabilitation consisted of reducing pain and effusion, improving quadriceps activation, achieving full knee extension range of motion and normal gait. Immediate mobilization and weight-bearing was applied unless contraindicated by the treating surgeon. The intermediate phase focused on motor control training and strengthening, and the advanced phase built up to plyometrics drills, power, and force development. All participants had to meet pre-determined criteria to pass to the following phase, and ultimately to be discharged and return to sport.[22,23]

During rehabilitation, the uninvolved limb was trained using bilateral exercises, and unilateral strength deficits were addressed according to patients' individual needs,

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as determined by the 6-week interval strength testing. The strength goal for all patients was LSI>90% and to achieve normative values according to their level of activity, based on the institution registry. For professional athletes the normative value from a matched control group was a minimum of 3 times peak torque to body weight.[39] Participants attended a minimum of 2 and up to 6 days per week depending on their activity level. On these days, stage and condition-appropriate sessions were provided by specialized sports physiotherapists, strength and conditioning coaches, hydrotherapists, and post-exercise recovery staff.

**Strength testing**

Strength testing was part of a routine assessment process done by an independent examiner. During each 6-week assessment, the patient first completed subjective questionnaires, and then was physically examined (range of motion, instrumented ligamentous laxity, etc.) before performing a warm-up of at least 6 minutes on a stationary bicycle. After warm-up, they undertook functional testing, based on the phase of the rehabilitation, followed by strength testing.

Isokinetic dynamometry was used for knee flexion and extension strength assessment.[5,6,36] The isokinetic strength test was performed concentrically, using a Biodex System 3 dynamometer (Biodex Medical Systems, NY) at 60°/s for 5 repetitions. The back rest was placed at an angle of 85°, and the participant was positioned in the chair fitted with two shoulder, one pelvic and one thigh strap. Both knee extension and knee flexion strength were measured throughout a range of motion of 90-0°, and participants were verbally encouraged to produce maximal force.[31] The uninvolved limb

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was tested first. Values for peak torque corrected for gravity were obtained. The parameters analyzed were quadriceps (knee extension) and hamstrings (knee flexion) isokinetic strength of the uninvolved limb, adjusted to body weight (Nm/kg) and representing the highest peak torque obtained during the test, and hamstring-to-quadriceps (H:Q) ratio.

The preoperative assessment time point chosen was the closest as possible to the surgery date, and postoperative assessments were performed at 1.5, 3, 4.5, 6, 7.5, and 9 months after surgery.

**Statistical Analysis**

Descriptive analyses of the participants' baseline characteristics were performed prior to inferential analyses. Linear mixed effects models were used with participants as random effects over the seven assessment time points, as this approach has been shown to better handle missing data.[21] Missing data were handled by using maximum likelihood estimation. The choice of the best model for each variable was made based on the Akaike information criterion. Fixed effects included: activity level category (professional and recreational), time after surgery, and time*activity level interactions. The parameter estimates for covariates (age and height, as fixed effects) were modelled, associated with patient characteristics. The final model's veracity was tested through examination of the distribution of the residuals. Specifically, visual examination of residuals distribution in comparison to a fitted normal, and inspection of the Q-Q plots, as well as conducting statistical tests of normality. Where a significant main effect or interaction was found, post-hoc testing with Tukey HSD adjustment for multiple

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comparisons was performed.[35] All data was analyzed with JMP Version 16.0, SAS Institute, and significance level was set at 0.05.

To account for the possible confounding of survivorship bias, considering those who were discharged earlier, and perhaps performing better, a sensitivity analysis was performed using time to discharge/RTS (4 categorizations: ≤6 months, >6 to ≤7.5 months, >7.5 to ≤9 months, and >9 months) as a fixed factor, which was not significant. Accordingly, the final analysis included the entire dataset.

## RESULTS

A total of 665 male participants were included in the analysis. Participant characteristics are described in **TABLE 1**. Pre-injury data were obtained from participants who were logistically able to attend assessment prior to their scheduled surgery (248 of 665).

Both professional and recreational athletes displayed a gradually increasing quadriceps and hamstrings strength at each time point during rehabilitation, from preoperative testing until 6 months after surgery, except for the recreational athlete group that exhibited a strength decrease at 6 weeks postoperative, with a subsequent increase at 3 months. For quadriceps strength, professional athletes significantly exceeded preoperative values by 3 months (2.99 Nm/kg, CI 95% 2.93-3.04, p=0.007), while recreational athletes overtook preoperative values by 6 months post-ACLR (2.77 Nm/kg, CI 95% 2.71-2.83, p<0.001). For hamstrings strength, both groups surpassed preoperative values by 4.5 months (1.71 Nm/kg, CI 95% 1.67-1.74, p<0.001 and 1.43 Nm/kg, CI 95% 1.40-1.46, p=0.002 for professional and recreational athletes

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respectively). For both activity level subgroups, most strength gains occurred between 1.5 and 4.5 months, increasing approximately 3-4% at each time point, and plateaued after 6 months (**TABLE 2** and **FIGURE 2**). At 9 months postoperative, professional athletes exhibited an increase of nearly 9% in quadriceps strength, and 10% in hamstrings strength on the uninvolved limb, while recreational athletes increased approximately 8% and 10% in quadriceps and hamstrings strength respectively, compared to the preoperative assessment. Concerning H:Q ratio, no significant differences were found between time points or comparing postoperative to preoperative values, with estimates ranging from 0.55-0.57 in professional and 0.52-0.53 in recreational athletes. Individual participant strength trajectory can be visualized in the **SUPPLEMENTAL APPENDIX, FIGURE 3**.

## DISCUSSION

Quadriceps and hamstrings strength of the uninvolved limb of male athletes progressively increased after ACLR. Professional athletes significantly exceeded preoperative values at 3 months for quadriceps and 4.5 months for hamstrings, while recreational athletes overtook pre-surgery strength values at 6 months for quadriceps and 4.5 months for hamstrings.

### The strength of the uninvolved limb does not decline after ACLR

Previous research evaluating uninvolved quadriceps and hamstrings muscle strength at multiple time points during rehabilitation have shown no changes between: preoperative and 6 months,[30] preoperative and 24 months,[34] 3 and 24 months,[4] and 1 and 6 months[10] post-ACLR, conflicting with our results that indicated a gradual increase

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at each time point from 1.5 to 6 months. Supporting our findings, two studies have reported increased muscle strength of the uninvolved limb between 4 to 12 weeks,[9] and from preoperative to 6 months postoperatively.[32] Additionally, a recent study demonstrated a strength increase between 6 and 9 months after surgery.[25]

We found a steeper increase in strength levels between 1.5 and 4.5 months, which started to plateau after 6 months. The rehabilitation program is strongly focused on strength development during this period, while after this stage there is a shift of focus towards more functional and sports related skills.[2,17] The average values between 6 and 9 months testing points may reflect an artifact of the remaining population, as those who performed best typically were discharged earlier, and therefore those that remained in rehabilitation at 9 months were likely the subset whose strength had not improved by as much.

Concerning H:Q ratio, the lack of change throughout rehabilitation indicates that hamstrings and quadriceps strength improved proportionately. H:Q ratio has been considered an important index of muscle strength imbalance, and although its predictive value of ACL or hamstrings injuries is still limited,[18] it has been associated with increased risk of reinjury.[22]

Comparison with the available literature is challenging as substantial methodological diversity is evident in terms of time between strength measurements, and total follow-up time. Moreover, rehabilitation programs' specifics such as if the uninvolved limb underwent strength training or not, are not detailed in these studies,[4,9,12,37] which makes it challenging to understand why differences in strength progression occur. Our results indicate that when adequate training stimulus is provided

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to the uninvolved leg during rehabilitation, patients probably will not display any deterioration in its strength, and given sufficient stimulus may even see an increase.

**Using preoperative measurements as baseline strength is not enough**

The fact that professional athletes have higher strength from the second post-operative assessment time point (1.5 months) compared to their preoperative values, may call into question whether there is already some detraining of the uninvolved limb post-injury and prior to surgery due to activity modification, as has been previously demonstrated.[8,11] This highlights the need to be cautious when relying on preoperative levels of knee strength,[43] and challenges the results from studies suggesting superiority of preoperative values over postoperative levels of strength to estimate risk of a secondary ACL injury.[43,45,47]

However, compared to pre-injury strength, preoperative values may be more easily available to clinicians if patients enroll in preoperative rehabilitation and represent an individual reference value that can be used to compare strength progression during rehabilitation.

**Beware of the "moving target"**

The argument that postoperative LSI may overestimate strength gains in the operated limb was not borne out in our data. A few studies have shown lower uninvolved limb strength after ACLR, when compared to the knee strength of matched-control participants.[13,14,26,38] Conversely, supporting the results of this study, two recent

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reviews showed that athletes with ACLR reach strength values comparable to healthy athletes after 7 months.[1,41]

Considering published normative data of knee muscle strength, the professional athletes in the present study have reached at 9 months nearly what is reported in the literature for a similar cohort, 3.23 Nm/kg for quadriceps strength (3.17 Nm/kg), and surpassed the 1.74 Nm/kg for hamstrings strength (1.77 Nm/kg).[39] For recreational athletes with a comparable activity level, normative data show values between 2.20-2.90 Nm/kg for knee extensors[4,15,16,41] and 1.46-1.57 Nm/kg for knee flexors[4,41] which are within the range of values of this study's results (for quadriceps 2.84 Nm/kg, and for hamstrings 1.51 Nm/kg). The differences in values found for our cohort could plausibly be explained by the activity level of these participants, as professional and recreational athletes are analyzed separately, unlike most studies looking at the uninvolved limb strength after ACLR, that report results in mixed athletic populations.

The progressive increase in the uninvolved limb strength throughout rehabilitation near to, and even surpassing, normative values shows that the uninvolved limb can be a "moving target". Although clinicians need to be aware of this when they calculate and interpret LSI, this metric can still be useful as a rehabilitation goal. Further research is required to determine if an optimal set of strength values exists in terms of performance and reinjury.

**Clinical implications**

Based on an increasing level of strength of the uninvolved limb in male athletes, reporting postoperative muscle strength symmetry in comparison to the uninvolved limb

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remains a practical measure after ACLR in this population as it can be measured at any time point and is subject specific. However, the limitations of these values need to be understood by practitioners and patients to better inform progress during rehabilitation. Preoperative strength values can be used if available, yet the timing of preoperative assessment should be considered to account for post-injury detraining that may affect comparisons of the uninvolved muscle strength values. Individual athlete-specific pre-injury strength levels, such as pre-season testing, would be better to provide patient-specific strength milestones, but if unavailable, normative values as reported in the literature or data from matched control individuals should be used for both professional and recreational male athletes, to inform strength progression in the uninvolved and involved limbs.

Through rehabilitation, systematic strength testing allows clinicians to follow a strength trajectory to inform clinical decisions, progression, and rehabilitation strategic planning, for exercise prescription and guidance throughout rehabilitation, and not only an end goal to achieve.

Rehabilitation programs can monitor and address the uninvolved limb strength in a controlled manner, while focusing on the operated knee, and should be applied according to the patients' goals, contextualized to their type and level of activity.

**Strengths and limitations**

The strength of these results was ensured as demographic, rehabilitation, and assessment variability were accounted for, by analyzing and presenting data from a cohort rehabilitated and assessed in a single center.

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Only a third of the participants were available for pre-surgery examination. The use of linear mixed models with maximum likelihood estimation is a robust method to handle missing data, and an additional analysis was performed to account for the possible confounding factor of survivorship bias.

Participants in this study followed an intensive standardized on-site rehabilitation protocol, which likely does not reflect usual care where this cost could be prohibitive (patients unable to attend rehabilitation sessions as frequently). The results were adjusted for patient-specific factors which provide better internal validity, but prevents extrapolation to other populations. The recruitment of only male participants from a single site suggests caution is required when interpreting these results and any comparison to other populations. These data cannot be generalized to female athletes, and less intensive or independently executed rehabilitation will likely display different strength results. A further limitation is that these changes have not been examined in comparison to changes in strength of the involved limb as this was not a primary aim of this research.

## CONCLUSION

Uninvolved limb quadriceps and hamstrings strength gradually improved after ACLR in male professional and recreational athletes undergoing intensive rehabilitation and surpassed preoperative values between 3 and 6 months post-surgery. Clinicians should be aware of the “moving target” nature of the uninvolved limb strength, which requires consistent monitoring during rehabilitation.

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## KEY POINTS

**Findings**: Uninvolved limb strength increased after ACLR to values greater than preoperative in this cohort of individuals undergoing an intensive rehabilitation program.

**Implications**: Strength LSI for quadriceps and hamstrings using the uninvolved limb as a standard, remains a practical criterion for strength assessment but clinicians should be mindful of the changing nature of the uninvolved limb's strength. Consistent measurements through rehabilitation may be useful to monitor progression and guide rehabilitation.

**Caution**: These data cannot generalize to female athletes nor those undertaking less intensive or unsupervised rehabilitation.

## STUDY DETAILS

**Authors' contributions:** OB, RK and VK conceived and designed the study. RK was responsible for overseeing data collection. VK and RW were responsible for the statistical analysis. OB drafted the manuscript. OB, RK, VK, RW and EK contributed to the critical review of the manuscript.

**Patient and Public involvement**: Patients and/or the public were not involved in the design, conduct, reporting, or dissemination plans of this research.

**Data sharing:** All data relevant to the study are included in the article or are available as supplementary material.

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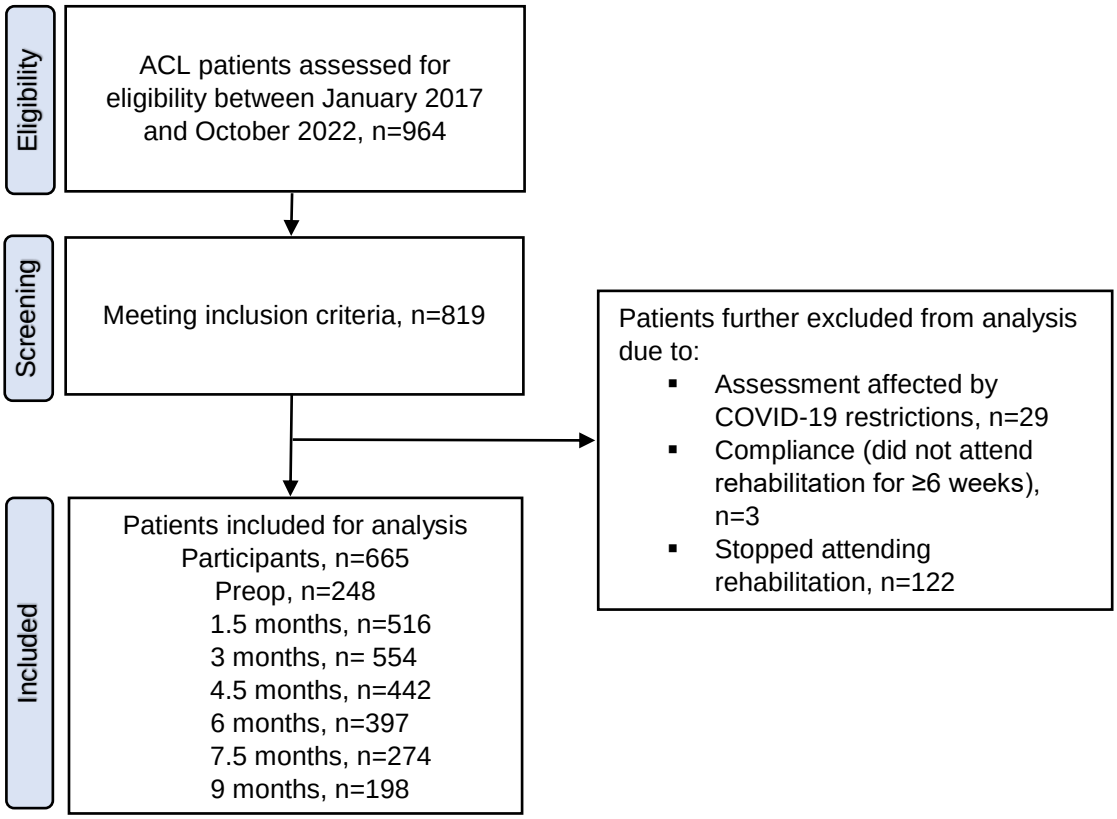


**FIGURE 1.** Flow chart for patient inclusion

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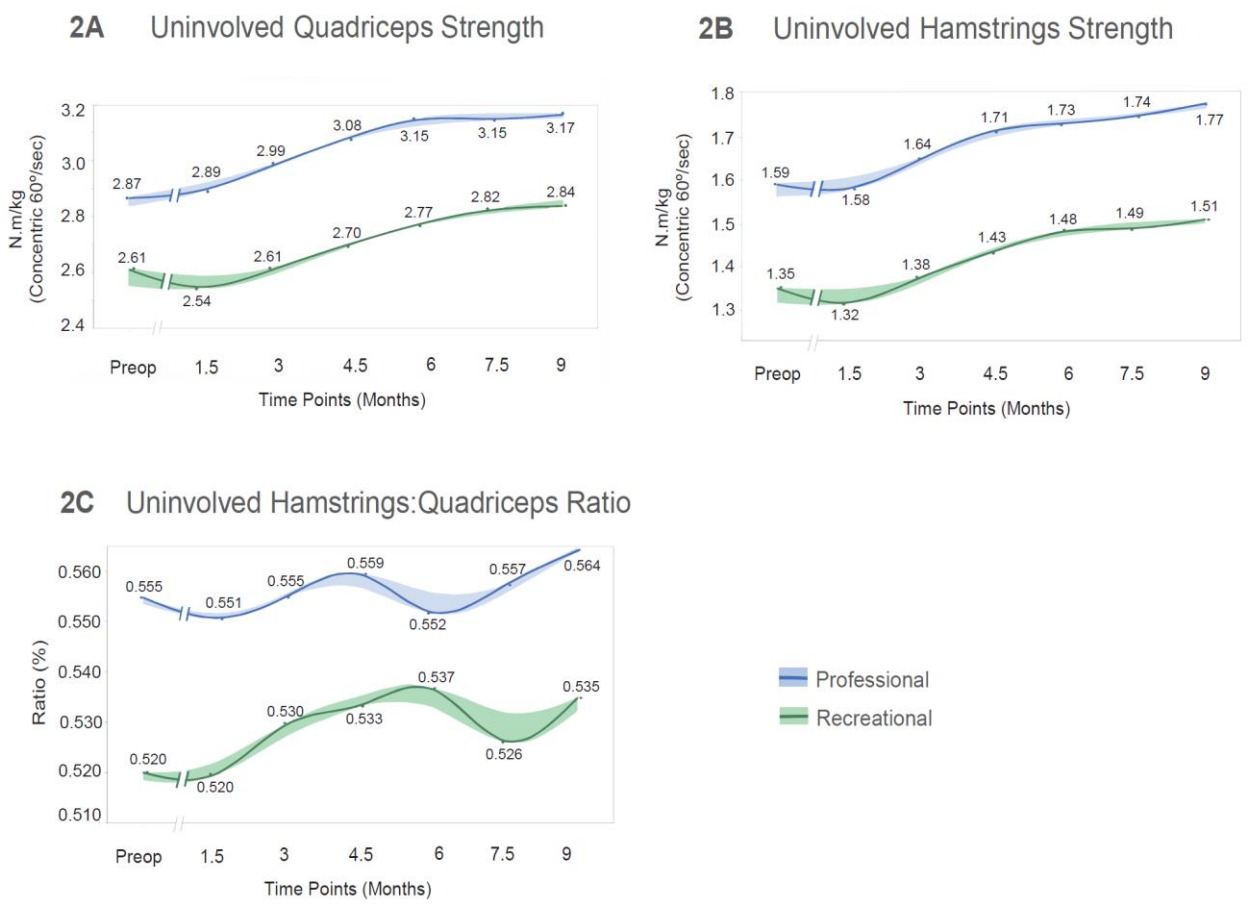


**FIGURE 2**. Uninvolved limb muscle strength progression during rehabilitation after ACLR in male professional and recreational athletes. The figure shows the strength estimates (adjusted for covariates) along with the line of best fit and the 95% confidence interval for this line for 2A: uninvolved quadriceps strength, 2B: uninvolved hamstrings strength, and 2C: Hamstrings:Quadriceps ratio. Note the scale of 2C ranges from 0.510 to 0.560, with each of the sub figures remaining within a relatively tight bound of less than 0.020, therefore an additional decimal place was added. Strength values were normalized to body weight (N.m/kg). H:Q ratio is expressed as %.
Abbreviations: Preop, preoperative.

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**TABLE 1**. Participant characteristics

| | Professional athletes (n=313) | | | Recreational athletes (n=352) | | |
|---|---|---|---|---|---|---|
| Age (years) | 23.9 | ± 5.3 | (15.0-45.0) | 29.2 | ± 7.8 | (15.4-54.3) |
| Body mass (kg) | 75.5 | ± 13.5 | (48.0-130.0) | 82.1 | ± 13.7 | (43.0-140.0) |
| Height (cm) | 177.3 | ± 8.2 | (157.0-211.0) | 174.2 | ± 6.9 | (154.0-195.0) |
| BMI (kg/m2) | 23.9 | ± 3.2 | (16.0-38.0) | 27.1 | ± 4.2 | (15.6-40.8) |
| Preinjury Tegner Score* | 9.2 | ± 0.4 | (Mdn=9, 9-10)* | 6.7 | ± 0.8 | (Mdn=7, 5-8)* |
| Time of preoperative testing to surgery (days)§ | 22.7 | ± 27.6 | (Mdn=12.5, 1-140, 24) | 33.7 | ± 40.1 | (Mdn=18, 0-197, 38) |
| Time of injury to preoperative testing (weeks)§ | 8.5 | ± 13.5 | (Mdn=3.0,0.1-64.0, 5.0) | 31.9 | ± 62.1 | (Mdn=9.9, 0.0-454.4, 22.8) |

Abbreviations: BMI, body mass index; Mdn, median.
All participants were male
Values are presented as mean±SD (minimum-maximum)
*Preinjury Tegner Score is presented as mean±SD (Median, minimum-maximum)
§Time of preoperative testing to surgery and Time of injury to preoperative testing are presented as mean±SD (Median, minimum-maximum, Interquartile range)

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**TABLE 2**. Strength values for quadriceps and hamstrings muscles and time point comparisons according to level of activity.

| | **Professional athletes** | | | | |
|---|---|---|---|---|---|
| | | Compared to Preop | | Compared to previous time point | |
| **QUAD** | | | | | |
| Time point | PT/BW (N.m/kg) | Mean Difference (95% CIs) | P-value | Mean Difference (95% CIs) | P-value |
| Preop | 2.92±0.52 | | | | |
| 1.5 | 2.96±0.50 | 0.03 (-0.07 to 0.12) | 1.00 | 0.03 (0.07 to 0.12) | 1.00 |
| 3 | 3.05±0.49 | 0.12 (0.02 to 0.22) | **0.007** | 0.09 (0.02 to 0.16) | **<0.001** |
| 4.5 | 3.15±0.46 | 0.21 (0.10 to 0.32) | **<0.001** | 0.09 (0.02 to 0.16) | **<0.001** |
| 6 | 3.20±0.49 | 0.28 (0.17 to 0.40) | **<0.001** | 0.07 (0.00 to 0.15) | 0.08 |
| 7.5 | 3.20±0.49 | 0.28 (0.16 to 0.41) | **<0.001** | 0.00 (-0.09 to 0.09) | 1.00 |
| 9 | 3.16±0.49 | 0.30 (0.16 to 0.44) | **<0.001** | 0.02 (-0.09 to 0.13) | 1.00 |
| **HAMS** | | | | | |
| Preop | 1.60±0.31 | | | | |
| 1.5 | 1.61±0.30 | -0.01 (-0.08 to 0.06) | 1.00 | -0.01 (-0.08 to 0.06) | 1.00 |
| 3 | 1.68±0.29 | 0.06 (-0.01 to 0.13) | 0.24 | 0.07 (0.02 to 0.12) | **<0.001** |
| 4.5 | 1.75±0.27 | 0.12 (0.05 to 0.19) | **<0.001** | 0.07 (0.02 to 0.11) | **<0.001** |
| 6 | 1.76±0.30 | 0.14 (0.07 to 0.21) | **<0.001** | 0.02 (-0.03 to 0.07) | 1.00 |
| 7.5 | 1.77±0.27 | 0.15 (0.08 to 0.23) | **<0.001** | 0.02 (-0.04 to 0.08) | 1.00 |
| 9 | 1.76±0.30 | 0.18 (0.10 to 0.27) | **<0.001** | 0.03 (-0.05 to 0.10) | 1.00 |
| **HQ Ratio** | Ratio (%) | Mean Difference (95% CIs) | P-value | Mean Difference (95% CIs) | P-value |
| Preop | 0.55±0.08 | | | | |

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| | | | | | |
|---|---|---|---|---|---|
| 1.5 | 0.55±0.09 | 0.00 (-0.03 to 0.02) | 1.00 | 0.00 (-0.03 to 0.02) | 1.00 |
| 3 | 0.56±0.08 | 0.00 (-0.02 to 0.02) | 1.00 | 0.00 (-0.01 to 0.02) | 1.00 |
| 4.5 | 0.56±0.07 | 0.00 (-0.02 to 0.03) | 1.00 | 0.00 (-0.01 to 0.02) | 1.00 |
| 6 | 0.55±0.08 | 0.00 (-0.03 to 0.02) | 1.00 | -0.01 (-0.02 to 0.01) | 0.95 |
| 7.5 | 0.56±0.07 | 0.00 (-0.02 to 0.03) | 1.00 | 0.01 (-0.01 to 0.02) | 1.00 |
| 9 | 0.56±0.08 | 0.01 (-0.02 to 0.04) | 1.00 | 0.01 (-0.02 to 0.03) | 1.00 |
| | **Recreational athletes** | | | | |
| | | Compared to Preop | | Compared to previous time point | |
| **QUAD** | | | | | |
| Time point | PT/BW (N.m/kg) | Mean Difference (95% CIs) | P-value | Mean Difference (95% CIs) | P-value |
| Preop | 2.59±0.47 | | | | |
| 1.5 | 2.52±0.48 | -0.07 (-0.15 to 0.02) | 0.33 | -0.07 (-0.15 to 0.02) | 0.33 |
| 3 | 2.57±0.48 | 0.00 (-0.09 to 0.09) | 1.00 | 0.07 (0.00 to 0.13) | **0.023** |
| 4.5 | 2.66±0.48 | 0.09 (-0.01 to 0.19) | 0.18 | 0.08 (0.02 to 0.15) | **0.004** |
| 6 | 2.76±0.49 | 0.16 (0.05 to 0.27) | **<0.001** | 0.07 (0.00 to 0.15) | 0.08 |
| 7.5 | 2.84±0.47 | 0.21 (0.10 to 0.33) | **<0.001** | 0.05 (-0.04 to 0.14) | 0.77 |
| 9 | 2.79±0.47 | 0.23 (0.09 to 0.36) | **<0.001** | 0.01 (-0.09 to 0.12) | 1.00 |
| **HAMS** | | | | | |
| Preop | 1.34±0.28 | | | | |
| 1.5 | 1.30±0.28 | -0.04 (-0.10 to 0.03) | 0.82 | -0.04 (-0.10 to 0.03) | 0.82 |
| 3 | 1.36±0.26 | 0.03 (-0.04 to 0.09) | 1.00 | 0.06 (0.02 to 0.11) | **0.001** |
| 4.5 | 1.42±0.29 | 0.08 (0.02 to 0.14) | **0.002** | 0.06 (0.01 to 0.10) | **0.002** |
| 6 | 1.48±0.28 | 0.13 (0.06 to 0.20) | **<0.001** | 0.05 (0.00 to 0.10) | 0.09 |
| 7.5 | 1.50±0.28 | 0.14 | **<0.001** | 0.01 | 1.00 |

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| | | | | | |
|---|---|---|---|---|---|
| | | (0.06 to 0.21) | | (-0.06 to 0.07) | |
| 9 | 1.48±0.27 | 0.16 (0.08 to 0.24) | **<0.001** | 0.02 (-0.05 to 0.09) | 1.00 |
| **HQ Ratio** | Ratio (%) | Mean Difference (95% CIs) | P-value | Mean Difference (95% CIs) | P-value |
| Preop | 0.52±0.07 | | | | |
| 1.5 | 0.52±0.07 | 0.00 (-0.02 to 0.02) | 1.00 | 0.00 (-0.02 to 0.02) | 1.00 |
| 3 | 0.53±0.07 | 0.01 (-0.01 to 0.03) | 0.95 | 0.01 (-0.01 to 0.03) | 0.64 |
| 4.5 | 0.54±0.07 | 0.01 (-0.01 to 0.04) | 0.73 | 0.00 (-0.01 to 0.02) | 1.00 |
| 6 | 0.54±0.07 | 0.02 (-0.01 to 0.04) | 0.46 | 0.00 (-0.01 to 0.02) | 1.00 |
| 7.5 | 0.53±0.07 | 0.01 (-0.02 to 0.03) | 1.00 | -0.01 (-0.03 to 0.01) | 0.85 |
| 9 | 0.54±0.08 | 0.02 (-0.01 to 0.04) | 0.84 | 0.01 (-0.01 to 0.03) | 1.00 |

Abbreviations: Quad, quadriceps; Hams, hamstrings; HQ, hamstrings-to-quadriceps; Preop, preoperative; CIs, confidence intervals; PT, peak torque; BW, body weight.
The uninvolved quadriceps and hamstrings strength is presented as unadjusted values (mean±SD) in N.m/kg and HQ ratio is presented as unadjusted values (mean±SD) of %, for both professional and recreational athletes. Comparisons between time points (to preoperative and to previous time point) show mean differences, 95% CIs, and p-values where bold indicates a statistically significant difference with $p<0.05$.

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## SUPPLEMENTAL APPENDIX

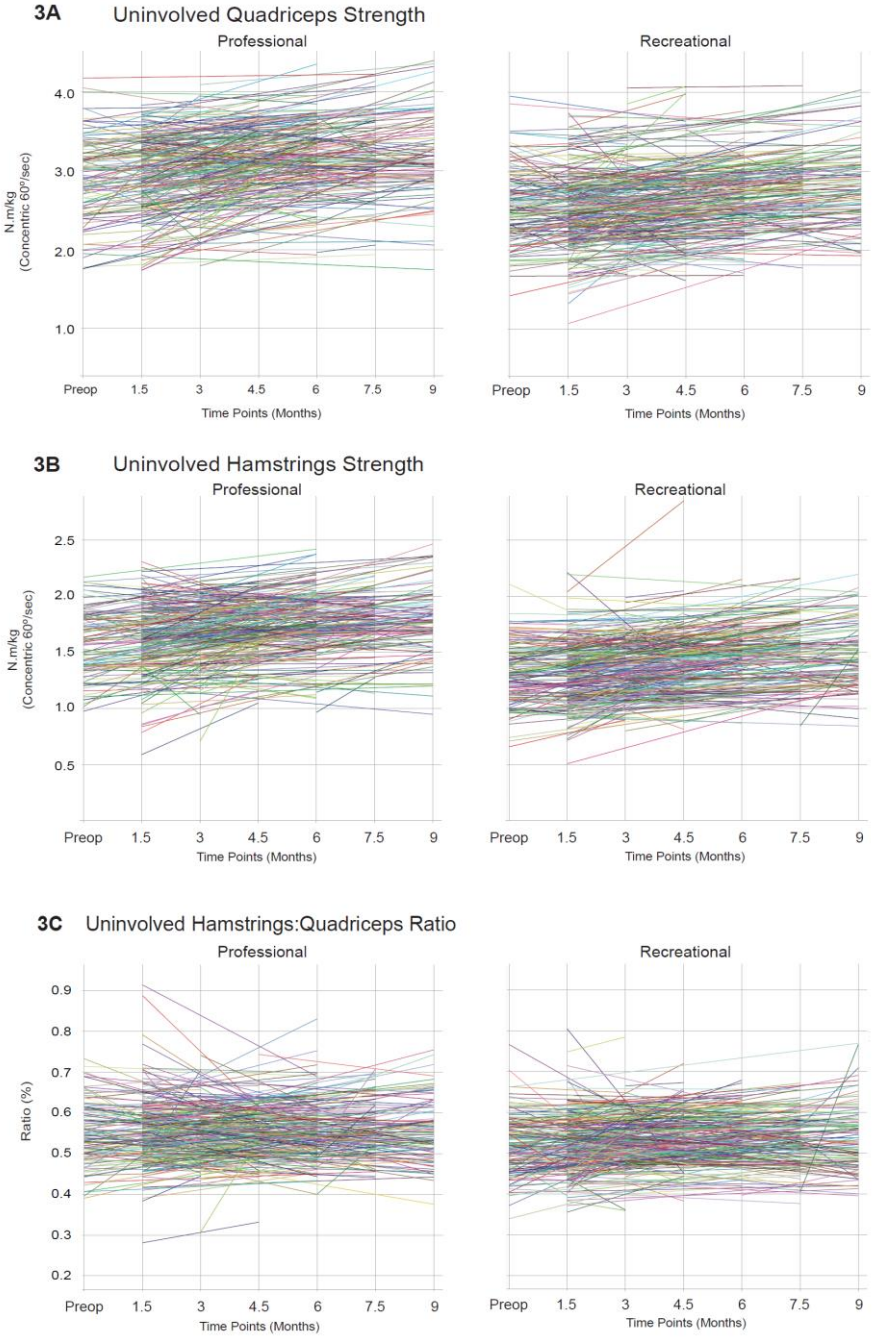


**FIGURE 3**. Uninvolved limb muscle strength progression during rehabilitation after ACLR in male professional and recreational athletes. Each line represents an individual participant's strength trajectory through each time point during rehabilitation for 3A: uninvolved quadriceps strength, 3B: uninvolved hamstrings strength, and 3C: Hamstrings:Quadriceps ratio.
Abbreviations: Preop, preoperative.

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